

Homework #2

Document Intelligence Project

Instructions

Prompt

In your Functional Brief, you mapped your function and flagged where AI could add value. Now you put one of those document-heavy workflows to work. You will apply the document-analysis techniques from AI.125, like extraction, summarization, classification, and cross-source synthesis, to source materials from your function. This builds on the Week 3–4 readings: Mollick’s “[Thinking Like an AI](#),” which explains why the model guesses plausibly and must be kept anchored to your sources, and OpenAI’s [Deep Research](#).

1. **Anchor to your function.** Choose 3–5 real source documents from your function for your company (e.g., HR policies, marketing briefs, vendor contracts, operational SOPs, or financial filings). Pick documents tied to one of the AI opportunities you identified in your Functional Brief, so the analysis feeds a real decision. List your sources at the top.
2. **Analyze each document and watch how the model behaves.** Use an LLM to extract key facts, summarize, and classify each source. As you go, apply the intuition from “Thinking Like an AI”: the model predicts plausible next words rather than retrieving from a database, so small changes in your prompt can change the output, and it can state false things confidently. Iterate your prompts, and keep the model anchored to the document.
3. **Verify and ground every key claim.** For each important finding, locate the exact passage in your source that supports it. Watch for two failure modes the readings flag: the model **pulling in outside or training knowledge even when you have given it a document**, and the model **over-attending to the start and end of a long prompt while missing the middle**. Run a second pass, ideally with a second model, to check your most important findings, and flag anything you could not verify.
4. **Synthesize across your sources.** Chain the analysis across all documents to surface common themes, contradictions, and gaps. This is the work that would normally take a junior analyst days by hand. Finally, use Deep Research (Week 4) to place your sources in a broader context or sanity-check a claim, but the synthesis must rest on your own documents, and you must verify any citations Deep Research returns.
5. **Produce the synthesis brief.** Write a concise multi-source synthesis brief (roughly 2–3 pages) that turns your analysis into actionable insights a manager in your function could use. **Connect those insights back to the function goal you identified in HW#1.**
6. **Document your process and governance.** Include an appendix with the key prompts you used and which phrasing worked, the tools and model tiers you used, how you verified outputs, and one data-governance consideration for using AI on these documents. For example, whether a confidential contract or HR file should be placed in a consumer tool at all. This continues the guardrails-versus-local-rules thinking from your Functional Brief.

The AI accelerates reading, but you are accountable for accuracy. As Mollick advises, using AI on material you understand well enough to catch its mistakes and verify anything you would be embarrassed to get wrong in front of your boss.

Submission Instructions

- Submit your synthesis brief (including the appendix) as a Google Doc on Campuswire.
- List your 3–5 source documents and cite them clearly.
- Label the appendix “Prompts, Tools & Verification,” so it is easy to find, and include your one data-governance note there.
- Late submissions follow the course Late Assignment Policy.

Grading Rubric

100 points total

#	Category & Target Behavior	Excellent	Good	Poor	Missing
1	Source Selection & Relevance (12 pts) <i>Chooses 3–5 documents tied to the function and to an opportunity from HW#1.</i>	Well-chosen, function-relevant sources clearly linked to a real decision.	Reasonable sources; link to HW#1 opportunity loose.	Sources thin, off-topic, or fewer than three.	No sources identified.
2	Document Analysis Technique (20 pts) <i>Applies extraction, summarization, and classification, iterating prompts as needed.</i>	Techniques used skillfully; prompt iteration evident and effective.	Techniques applied; some shallow or little iteration.	Minimal or incorrect technique use.	No analysis performed.
3	Cross-Source Synthesis (25 pts) <i>Identifies themes, contradictions, and gaps; uses Deep Research appropriately if at all.</i>	Insightful synthesis connecting sources; any Deep Research use is supplementary and verified.	Synthesis present; partly surface-level.	Little true synthesis; documents treated in isolation.	No synthesis.
4	Grounding & Verification (18 pts) <i>Anchors key claims to source passages and runs a second-pass / second-model check; aware of hallucination.</i>	Claims traced to sources, verified, and unverifiable items flagged.	Some grounding and verification; uneven.	Little grounding; takes AI output at face value.	No verification attempted.
5	Actionable Insight Quality (15 pts) <i>Turns analysis into decision-ready insights aligned to the function goal.</i>	Clear, specific, decision-ready insights tied to function goals.	Useful insights; some vague or loosely linked.	Insights generic or unsupported.	No insights offered.
6	Process & Governance Notes (10 pts)	Transparent, complete appendix with a	Most elements present;	Sparse appendix; no	No appendix.

#	Category & Target Behavior	Excellent	Good	Poor	Missing
	<i>Appendix documents prompts, tools/model tiers, verification method, and one governance consideration.</i>	thoughtful governance note.	governance note thin.	governance note.	